

Detection of Crop Diseases Using UAV-Captured Images

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Abstract: This work investigates the detection and classification of agricultural diseases using machine learning approaches based on UAV-captured photos. In particular, plant leaf photos taken by UAVs were processed to extract HOG (Histogram of Oriented Gradients) features, which were then fed into a logistic regression classifier. Dimensionality reduction was accomplished using Principal Component Analysis (PCA), while unsupervised learning was accomplished using K-Means clustering. With a classification accuracy of 88.02%, the method shows promise for crop health monitoring using UAVs in real time.

1. INTRODUCTION

In order to maintain the global economy and provide food security for billions of people, agriculture is essential. However, the emergence of plant diseases is one of the most enduring problems that farmers and agricultural researchers must deal with. These illnesses can cause large output losses if they are not identified and treated in a timely manner, impacting local farming communities as well as the global food supply chain. Therefore, one of the most important steps in reducing these losses is the timely detection of crop diseases.

Disease identification has historically depended on agricultural specialists performing manual inspections, which is labor-intensive, time-consuming, and frequently unfeasible for large-scale farming operations. Furthermore, without professional assistance, a correct diagnosis can be challenging because early-stage symptoms can be mild and differ between crop species.

Precision agriculture has seen a shift in recent years due to developments in machine learning and computer vision. Researchers may now create automated systems that can accurately identify diseases because to the availability of high-resolution photos and strong analytical tools. Unmanned Aerial Vehicles (UAVs) are one of the transformative technologies supporting this shift. UAVs, often known as drones, are able to swiftly and affordably take aerial photos of large agricultural areas. UAVs offer a scalable crop health monitoring option when outfitted with the right imaging and processing equipment.

With an emphasis on leaf-level indicators like discolouration, spots, and textural changes, this study investigates a machine learning method for identifying plant diseases using UAV-captured photos. The system makes use of traditional computer vision methods, such as logistic regression for classification and histogram of oriented gradients (HOG) for feature extraction. Our methodology offers a lightweight yet efficient solution appropriate for implementation in real-time UAV platforms, in contrast to deep learning techniques that necessitate substantial datasets and computational resources.

This paper's main contributions are:

- creation of a traditional machine learning pipeline that uses UAV photos to diagnose diseases.
- High-dimensional feature vectors are handled by applying dimensionality reduction techniques.
- The model is tested on a publicly accessible dataset of plant diseases, and the results are displayed via confusion and clustering matrices.

Practical ramifications, difficulties, and prospects for UAV-based disease surveillance are discussed.

This project is to support the development of autonomous, real-time crop health monitoring systems that provide farmers with timely and relevant insights by fusing UAV technology with effective machine learning models.

2. RELATED WORK

In recent years, the nexus between artificial intelligence and agriculture has attracted a lot of scientific interest, especially in the area of automated crop disease identification. Several studies have suggested ways to accurately diagnose and categorize plant diseases using deep learning, machine learning, and image processing approaches. The pertinent literature is reviewed in this part, which also places our strategy within the framework of previous studies.

To get over the drawbacks of manually created feature-based methods, researchers started using machine learning models. For the extraction of robust features, the Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and Local Binary Patterns (LBP) gained popularity. Classifiers like k-Nearest Neighbors (k-NN), SVM, Decision Trees, and Logistic Regression were then fed these features.

Ferentinos (2018) used the PlantVillage dataset to apply a classification framework that employed several machine learning algorithms, resulting in a high degree of disease identification accuracy. Despite being less computationally demanding than deep learning models, the effectiveness of these traditional machine learning techniques frequently depends significantly on the caliber of feature extraction.

Our disease detection system's fundamental component is a traditional computer vision pipeline that recognizes diseased crop leaves by processing photos taken by unmanned aerial vehicles. We employ a lightweight, interpretable, and effective machine learning approach that makes use of handcrafted features for reliable performance under resource restrictions rather than depending on computationally costly deep learning architectures.

Using HOG for Feature Extraction
Finding useful information from unprocessed images that may be used for classification is known as feature extraction, and it is one of the most important

processes in computer vision. The Histogram of Oriented Gradients (HOG), a proven method for identifying texture and shape patterns, is used in our approach.

In order to capture edge and structure information that frequently corresponds with outwardly obvious illness symptoms including spots, lesions, and color changes, HOG computes gradient orientations across specific regions of an image.

To extract a high-dimensional feature vector for every image, the images are transformed to grayscale, scaled for uniformity, and then run through the HOG descriptor.

One benefit of this approach is its low computing overhead, which makes it appropriate for embedded or real-time systems like UAVs.

3. METHODOLOGY

Using computer vision algorithms on photos taken by Unmanned Aerial Vehicles (UAVs), the suggested system seeks to identify crop diseases. The overall architecture, dataset used, preprocessing procedures, detection model, and intended alarm system for real-time or near-real-time deployment in agricultural fields are described in this part.

3.1. System Overview

The following steps make up the system's modular pipeline design:

Image Acquisition: Top-down photos of crop canopies are taken by UAVs fitted with RGB cameras as they fly over agricultural areas.

Data handling: To improve quality and consistency, captured photos are preprocessed before being supplied into the system.

Feature Extraction & Classification: The photos are classified into illness groups using traditional machine learning approaches.

Alert Mechanism: To notify farmers or agronomists of illnesses found, a notification mechanism might be included.

Because of its lightweight design, this pipeline can be implemented for real-time use on edge devices or UAV-based processors.

3.2. Dataset:

Tens of thousands of tagged plant leaf photos taken in regulated lighting settings make up the publicly accessible PlantVillage dataset, which we use as a subset.

Three distinct illness classes were chosen for this investigation:

Spot of Bacteria in Pepper Bell

Late Blight in Potatoes

The late blight of tomatoes

The disease class is used to identify each image. A fair representation across classes is ensured by the 3906 photos utilized in total in this study. The initial dimensions of each image were 256×256 pixels.

3.3. Preprocessing:

Several preprocessing procedures are used prior to feature extraction and classification:

Image Resizing: All images are downsized to 128×128 pixels in order to preserve consistency and lower computing costs.

Color Conversion: Since the selected feature descriptor (HOG) performs well with single-channel data, images are transformed from RGB to grayscale.

Data Shuffling: To prevent bias in the training and testing sets, the dataset is randomized.

Label Encoding: Label Encoding transforms string labels, such as class names, into numerical form.

By taking these actions, the model's generalization is enhanced and consistency is guaranteed.

3.4.Alert System:

An alarm system is envisioned in order to make this system feasible for agricultural deployment. When the system finds a sick plant, it can:

Send out a real-time alarm (by onboard signal, SMS, or app notification).

Use UAV GPS metadata to mark the affected area's

geolocation.

Agronomists can use an updated dashboard to see hotspots and plan focused actions.

This characteristic minimizes crop losses and increases productivity by enabling early reaction and greatly reducing the spread of diseases.

4. Experiments and Results:

The experimental setup, test cases, and outcomes of the suggested system's implementation are shown in this part. The purpose of the research was to confirm that agricultural diseases could be accurately identified from UAV-captured photos using traditional machine learning methods, specifically HOG-based feature extraction.

4.1. Experimental Setup:

Several open-source tools, including OpenCV, Scikit-learn, NumPy, and Matplotlib, were used in the Python experiments, which were conducted in a Kaggle Notebook environment.

Hardware: Kaggle, a cloud-based environment with GPU support if necessary

OpenCV, NumPy, scikit-learn, matplotlib, seaborn, pandas, and PIL are examples of software libraries.

Dimensions of the image: resized to 128 by 128 pixels

Histogram of Oriented Gradients (HOG) for Feature Extraction

A feature Before PCA, each image had a vector length of about 34,020.

Dimensionality Reduction: Features were broken down into 2,100 components using Principal Component Analysis (PCA).

Classifier Logistic Regression with a maximum of 1000 iterations

Evaluation metrics include confusion matrix, F1-score, recall, accuracy, and precision.

Using stratified sampling, the dataset was divided into 75% training and 25% testing.

4.2. Test Cases:

We chose pictures from the following three dataset types to make sure the model works well across them:

Bacterial Spot in the Pepper Bell

Late Blight in Potatoes

The late blight of tomatoes

To assess model generalization, test instances were selected at random from the dataset. The entire pipeline (preprocessing → HOG extraction → PCA → classification) was applied to every test image, and predictions and actual labels were compared.

Multiple samples were plotted with expected and true labels to manually confirm accuracy for improved visual comprehension.



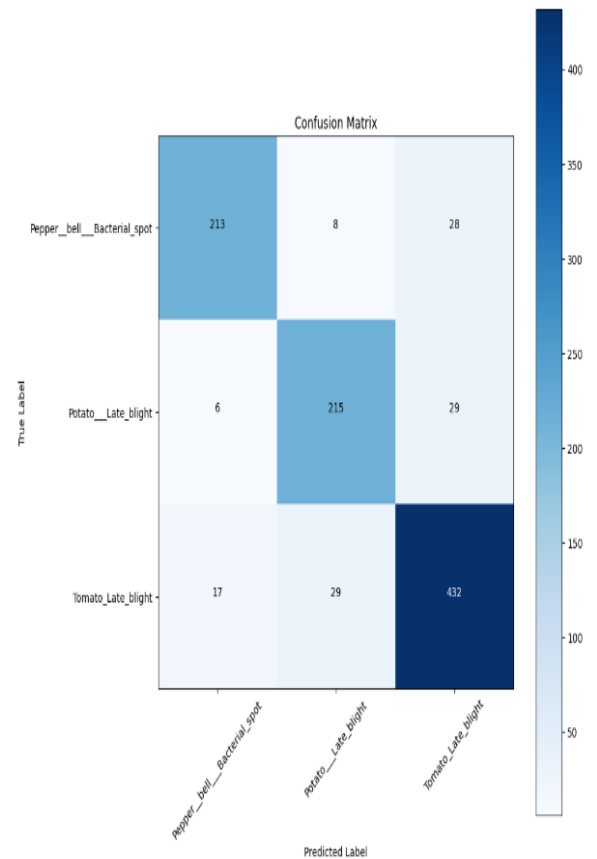
4.3. Results Analysis:

The trained Logistic Regression model achieved an overall **accuracy of 88.02%**. Below is a breakdown of the performance by class:

Class	Precision	Recall	F1-score	Support
Pepper Bell – Bacterial Spot	0.90	0.86	0.88	249
Potato – Late Blight	0.85	0.86	0.86	250
Tomato – Late Blight	0.88	0.90	0.89	478

Overall Accuracy: 88.02%

These results indicate a strong ability of the model to distinguish between similar diseases affecting different crops.



5. Discussion:

A more thorough examination of the advantages, difficulties, and applications of the suggested method for identifying agricultural diseases using UAV-captured photos is given in this section. In order to evaluate the system's practicality in real-world agricultural environments, the conversation considers both its technical and practical features.

5.1. Strength:

High Accuracy with Lightweight Model: The model's accuracy of over 88%, which is competitive for multiclass classification in plant disease detection, was attained while utilizing a traditional machine learning pipeline (HOG + PCA + Logistic Regression). This demonstrates that deep learning is not necessary for very effective feature extraction techniques.

Effective Feature Extraction: The model was able to extract crucial texture and shape information from the leaf photos by utilizing the Histogram of Oriented Gradients (HOG). Compared to deep convolutional feature maps, this approach is more interpretable and computationally less expensive.

Modular and Scalable Pipeline: A scikit-learn pipeline was used to construct the entire procedure, from preprocessing to classification. This facilitates the extension, updating, and replacement of specific components of the technique (e.g., substituting Random Forest or SVM for Logistic Regression).

Use of UAV Imagery in Practice: Using UAV-captured photos allows field-level disease monitoring to scale. Without the need for human assistance, UAVs can swiftly cover wide regions and take detailed pictures, enabling high-resolution crop analysis.

Interpretability and Debugging: Compared to black-box models like deep neural networks, traditional machine learning techniques like PCA and logistic regression make it easier for researchers to see feature importance, interpret decisions, and debug.

5.2. Challenges:

Visual Similarity Between Diseases: It might be challenging to differentiate certain plant diseases using simple texture descriptors alone because they can present with remarkably similar symptoms (such as leaf blight on various crops). Classification errors

result from this, particularly when comparing classes such as "Tomato Late Blight" and "Potato Late Blight."

Restricted Dataset Classes: Only three classes were used to train the model, which restricts how broadly it may be applied. Larger and more varied datasets are needed for real-world scenarios, which encompass a wider range of illnesses and symptoms.

Lighting and Environmental Variability: The time of day, the weather, and background noise can all have a significant impact on UAV photos. The consistency of feature extraction may be adversely affected by this unpredictability, particularly when employing grayscale or edge-based techniques such as HOG.

High-dimensional features have a computational cost because, despite their effectiveness, HOG generates a lot of features (around 34,000 per image), which lengthens computation times and uses more memory. Prior to classification, dimensionality reduction was necessary, which made the pipeline more complicated.

5.3. Solution and Limitations:

Use of Principal Component Analysis (PCA) for Dimensionality Reduction: PCA was employed to address the high dimensionality of HOG features. By decreasing the feature space to 2,100 dimensions, PCA improved training time, decreased overfitting, and preserved important information.

Train-Test Split with Stratified Sampling: A more accurate and balanced assessment resulted from stratified sampling, which made sure that the class distribution was maintained between training and testing datasets.

Limitations in Real-time Deployment: Although the existing model works well in an offline configuration, it would need to be further optimized and perhaps include embedded processing capabilities in order to be deployed in real-time on UAVs.

These issues are not yet covered in this study.

Lack of Deep Learning Comparison: Although the study's traditional methods were effective, comparisons to CNN-based models for benchmarking would give the results greater significance.

5.4. Practical Implications:

Early Disease Detection and Intervention: The system may be a helpful tool for early disease detection, allowing farmers to take prompt action and minimizing crop losses.

Cost-Effective Monitoring: By reducing reliance on pricey and sophisticated deep learning technology, using UAVs for aerial picture acquisition and traditional machine learning for disease diagnosis makes it more affordable for small-scale farms.

Integration with Agricultural IoT Systems: For comprehensive farm management, the disease detection model can be incorporated into larger smart farming systems that also include yield prediction, irrigation control, and pest detection.

Potential for Expansion: With the right retraining on pertinent datasets, the pipeline's modular design enables it to scale across many crop types, diseases, and geographical areas.

6. Conclusion and future work:

This research presents an effective and scalable approach to crop disease detection using images captured by Unmanned Aerial Vehicles (UAVs). By leveraging classical computer vision and machine learning techniques, the system identifies diseases in crops such as pepper, potato, and tomato with considerable accuracy. The use of Histogram of Oriented Gradients (HOG) for feature extraction, followed by dimensionality reduction via Principal Component Analysis (PCA), and classification using Logistic Regression, proved to be a robust combination. The model achieved an accuracy of approximately 88%, demonstrating the feasibility of using lightweight, interpretable models for real-world agricultural applications.

The overall pipeline is modular, easy to understand, and computationally efficient, making it suitable for deployment in scenarios where resources may be limited. Additionally, the integration of UAVs significantly enhances scalability, allowing coverage of large farmlands with minimal human intervention. This can lead to earlier disease detection, reduced pesticide usage, and ultimately, increased crop yield and food security.

6.1. Future work:

Even though the suggested approach has produced encouraging outcomes, there are a number of ways to improve it going forward:

Integration of Deep Learning Models: Adding Transformer-based designs or Convolutional Neural Networks (CNNs) could increase accuracy, particularly for more complicated datasets or closely related illness types. For the purpose of detecting plant diseases, pre-trained models such as ResNet, EfficientNet, or ViT can be refined.

Multi-Disease and Multi-Crop Detection: More crops and disease classes should be supported by future versions. This would entail selecting a dataset that is more balanced and diversified, as well as maybe applying hierarchical categorization techniques.

UAV Real-Time Deployment: By refining the model for real-time inference on edge devices connected to UAVs (like the NVIDIA Jetson Nano), farmers will be able to receive prompt feedback in the field and take prompt action.

Environmental Robustness: To deal with variations in lighting, weather, and occlusions (such as overlapping leaves or shadows), future research should investigate picture improvement approaches and data augmentation procedures.

Geotagging and Disease Mapping: Farmers and agronomists can track the patterns of disease spread over time by using GPS data and UAV imagery to map disease occurrences spatially.

Integration with IoT and Decision Support Systems: For more comprehensive decision-making, the model might be integrated into a broader smart farming ecosystem that also includes soil analysis, irrigation control, and weather monitoring.

Farmer-Centric Mobile Application: An easy-to-use online or mobile application may be created that would enable farmers to submit pictures and get prompt disease diagnosis and treatment recommendations.

To sum up, this study establishes the foundation for a crop disease detection system that uses UAV imagery and is scalable, interpretable, and effective. Such systems have the potential to transform precision agriculture and increase the efficiency of food production globally with further development.

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